PHY101: Introduction to Physics-1

Monsoon, 2018

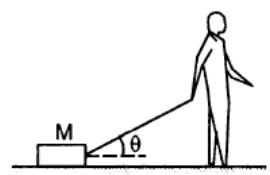
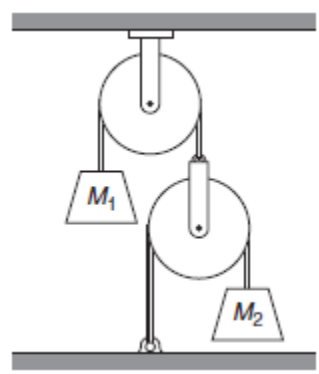
Tutorial-4

1. A particle moves outward along a spiral. Its trajectory is given by $r = A\theta$, where A is a constant. θ increases with time according to $\theta = \alpha t^2/2$ where α is another constant. Show that the radial acceleration is zero when $\theta = 1/\sqrt{2}$ rad.

2. A block is given an initial velocity of 5.00 m/s up a frictionless 20.0° incline. How far up the incline does the block slide before coming to rest?

3. Besides its weight, a 2.8 kg object is subject to one other constant force. The object starts from rest and in 1.2 s experiences a displacement of $(4.2\,\hat{\imath} - 3.3\hat{\jmath})$ m, where the direction $\hat{\jmath}$ is the upward vertical direction. Determine the other force.

4. A block of mass M is pulled on a smooth horizontal table by a string making an angle θ with the horizontal as shown in the figure below. If the acceleration of the block is a, find the force applied by the string and by the table on the block.



5. Masses M_1 and M_2 are connected to a system of strings and pulleys as shown in the figure. The strings are massless and inextensible, and the pulleys are massless and frictionless. Find the acceleration of M_1.